

Secure Exit



SIMATS Online Programming Lab

JAVA PROGRAMMING



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QUESTIONS

Q1

Banking Transaction System –
Exception Handling

10 marks

Q2

Hospital Patient Registration –
User-Defined Exception

10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions

10 marks

Q4

Hospital Emergency System –
Thread Priority

10 marks

Q5

Online Food Delivery – Multiple
Threads and Execution Order

10 marks

Q1 Banking Transaction System – Exception Handling

10 marks

Develop a Java program for an ATM withdrawal system. The program accepts the account balance and withdrawal amount.

The program must:

- Accept the current account balance and withdrawal amount.
- Throw a user-defined `InsufficientBalanceException` when the withdrawal amount exceeds the balance.
- Handle withdrawal amounts that are zero or negative.
- Handle invalid numeric input using a suitable built-in exception.
- Display a meaningful transaction result.
- Use try-catch-finally so that the transaction completion message is always displayed.

Sample Test Cases:

Input / Scenario Expected Result Purpose

10000 3000 Normal case

10000 12000 Boundary / exception case

10000 -500 Invalid input

abc 3000 Invalid input

5/5 answered

Your Code

```

1 class Main
2 {
3     public static void main(String []args)
4     {
5         double balance =10000,amount =3000;
6         try
7         {
8             if(amount<=0)throw new IllegalArgumentException();
9             if(amount>balance)throw new InsufficientBalanceException();
10            System.out.println("Withdrawal successful");
11        }
12        catch(Exception e)
13        {
14            System.out.println(e.getMessage());
15        }
16        finally
17        {
18            System.out.println("Transaction completed");
19        }
20    }
21 }
22 class InsufficientBalanceException extends Exception
23 {
24     InsufficientBalanceException(String s)
25     {
26         super(s);
27     }
28 }

```

Input

stdin input

Output

Withdrawal successful
Transaction completed

✓ Submitted

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